

Why Workers Stay: Pay, Beliefs, and Attachment

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Motivation

Job-to-job mobility is important for wage growth & labor market dynamism

Many reasons workers may not search or switch firms

- ▶ Beliefs about the distribution of pay (average outside option)
- ▶ Beliefs about who pays what
- ▶ Beliefs about offer probabilities
- ▶ Switching costs/inertia

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What the key friction to mobility is matters for policy & theory

- ▶ Variety of policy interventions directly target worker information (Council of Economic Advisors, 2016)
- ▶ Switching costs suggest interventions may need to occur earlier
- ▶ Beliefs about pay at the core of directed vs. random search models

This Paper

1. What do workers believe about firm pay?
2. Do these beliefs affect workers' search behavior?
3. Why don't workers move?

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- ▶ Surveyed >10k full-time German workers through the IAB.
- ▶ Developed an infrastructure that allows us to link workers' answers about specific, named firms to data, incl. admin data on those firms.
- ▶ Asked workers what they think they would make at specific outside firms.

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- ▶ Asked workers what firms they would apply to if they wanted to switch firms.
- ▶ Asked workers whether they would apply to specific researcher-provided firms.

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- ▶ Asked workers how likely they would be to search if outside pay were X% higher.
- ▶ Firm-specific hypothetical choice experiments: rank hypothetical **offers** with randomized raises.

Preview of Results

1. Workers do not believe in a uniform outside option.

- ▶ About half of workers report knowing wages at time of application.
- ▶ Workers expect their wages would vary across firms, holding position fixed.
- ▶ There are "firm effects" in workers' beliefs.
- ▶ Expected wages and wage premia are correlated with admin data predictions.

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- ▶ Joint test that workers direct their search and believe in firm rents.
- ▶ Amenity valuations are positively correlated with perceived firm wage premia.

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3. Many workers are not marginal to their current firm.

- ▶ Switching costs are 7-18% of a worker's annual salary.
- ▶ Amenity valuations differ between insiders and outsiders.

Related Literature

Ethnographic Research on the Labor Market

- ▶ e.g. Myers and Shultz (1951); Reynolds (1951); Rees and Shultz (1970)
- ▶ This paper: compare workers' preferences and beliefs to admin. data

Firm Wage Effects and Amenities

- ▶ e.g. Abowd et al. (1999); Card et al. (2013); Song et al. (2019); Rosen (1986); Sorkin (2018)
- ▶ This paper: workers expect firms to vary in wages and amenities, and believe $\text{cov}(a_j, \psi_j) > 0$

Workers' Information About the Labor Market

- ▶ e.g. Reynolds (1951); Caldwell and Harmon (2019); Jäger et al. (2024); Cullen (2023)
- ▶ This paper: a large share of workers have information on what specific outside firms would pay

Directed Search and Queuing

- ▶ e.g. Holzer et al. (1991); Banfi and Villena-Roldan (2019); Marinescu and Wolthoff (2020); Belot et al. (2022); He et al. (2023)
- ▶ This paper: workers are more likely to consider high wage firms

Framework

Model Overview

- ▶ Workers can apply to multiple firms from $\mathcal{J} = \{1, 2, \dots, J\}$
- ▶ Application costs κ
- ▶ Offer probabilities p_k
- ▶ Switching cost s_i if receive and accept an offer from a new firm
- ▶ Discount rate $\delta \in (0, 1)$, exogenous job destruction rate λ

Search is **directed** if there exists at least one pair of firms (k, l) with $\tilde{w}_{ikt} \neq \tilde{w}_{ilt}$ such that altering these beliefs, while holding aggregate distributions constant, changes the worker's application decisions.

Empirical Predictions

1. $P(\text{Search})$ is increasing (decreasing) in workers' perceived outside options (switching costs)
2. Inefficient non-search is less common when workers have firm-specific information
3. Directed searchers apply more to firms with higher perceived wages
4. The application-wage elasticity is increasing in the share of directed searchers.
 - ▶ Important for wage-setting in a wide class of models
5. $P(\text{Move})$ is increasing (decreasing) in workers' perceived outside options (switching costs)

Data

Worker Survey and Linkages

Fielded a survey to full-time German workers between 25 and 50

- ▶ Invitations sent from the IAB by mail; survey completed online
- ▶ Initial survey (11.4% response rate) conducted in 2022 
- ▶ Follow-up survey (51% response rate) conducted in spring 2024 
- ▶ Randomized incentives to account for selection into non-response 
- ▶ Over-sampled workers at firms surveyed in Caldwell, Haegele and Heining (2024)

Linked the ~10k worker-level responses to IAB Social Security records

- ▶ Worker histories, occupation and industry codes, AKM effects (Bellmann et al., 2020)

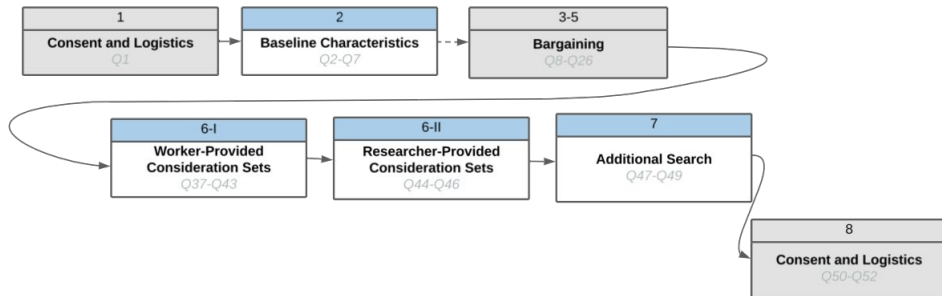
Constructed a novel linkage between workers' responses about specific firms and:

- ▶ Social Security records and wage premia associated with those firms
- ▶ ORBIS and Kununu 

Descriptive Statistics

	<u>Initial Wave</u>		<u>Initial and Follow-Up</u>	
	Mean	Std. Dev.	Mean	Std. Dev.
	(1)	(2)	(3)	(4)
<u>Demographics</u>				
Female	0.40	(0.49)	0.37	(0.48)
Age	31.13	(5.18)	31.31	(5.19)
German Citizen	0.89	(0.32)	0.92	(0.28)
College Degree	0.53	(0.50)	0.61	(0.49)
Apprenticeship	0.37	(0.48)	0.31	(0.46)
<u>Employment</u>				
Daily Pay (Imputed)	136.06	(47.81)	143.03	(47.66)
Censored Pay	0.06	(0.24)	0.07	(0.25)
Hours (Survey)	40.36	(6.47)	40.43	(5.90)
CBA Covered (Survey)	0.48	(0.50)	0.45	(0.50)
Manufacturing Sector	0.22	(0.41)	0.23	(0.42)
Retail Sector	0.09	(0.29)	0.09	(0.29)
Professional Sector	0.15	(0.36)	0.17	(0.37)
Observations	9756		3575	

Questions About Two Types of Firms

[▶ Initial Survey Detail](#)

Selecting Researcher-Provided Firms

1. Not too many firms

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3. Known to respondents
4. Horizontally reasonable 
5. Vertically reasonable

Selecting Researcher-Provided Firms

1. Not too many firms
 - ▶ 30 well-known German firms: 18 publicly listed, 12 family-owned.
 - ▶ Received more than 39.1 million page views on employer rating platform Kununu.
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 - ▶ 17 of the 30 firms are among the top 100 firms that workers named themselves.
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- ▶ Occupational distributions match
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4. Horizontally reasonable ▶
 - ▶ Occupational distributions match
5. Vertically reasonable
 - ▶ Median and mean pay throughout observed worker distribution ▶

1. Eliciting Information About Search

Suppose you planned to move to a new company in the next {one/three/six} months. Would you consider applying to any of these? Please select all that apply.

- ☐ Company 1
- ☐ Company 2
- ☐ Company 3
- ☐ Company 4
- ☐ Company 5
- ☐ Company 6
- ☐ Company 7
- ☐ I would not apply to any of these

2. Eliciting Information About Wages

Suppose you planned to move to a new company in the next {one/three/six} months. Would you consider applying to any of these? Please select all that apply.

- ☐ Company 1
- ☐ Company 2
- ☐ Company 3
- ☐ Company 4
- ☐ Company 5
- ☐ Company 6
- ☐ Company 7
- ☐ I would not apply to any of these

What do you think your gross annual pay would be if you worked at these companies in a position similar to your current one?

Company 2: [Fill in gross pay]

Company 4: [Fill in gross pay]

Company 7: [Fill in gross pay]

3. Eliciting Preferences via Hypothetical Choice Experiments

Suppose you planned to move to a new company in the next {one/three/six} months. Would you consider applying to any of these? Please select all that apply.

- ☐ Company 1
- ☐ Company 2
- ☐ Company 3
- ☐ Company 4
- ☐ Company 5
- ☐ Company 6
- ☐ Company 7
- ☐ I would not apply to any of these

What do you think your gross annual pay would be if you worked at these companies in a position similar to your current one?

Company 2: [Fill in gross pay]

Company 4: [Fill in gross pay]

Company 7: [Fill in gross pay]

Suppose you can remain at your current company or switch to any of the companies listed below and immediately receive the raise specified.

Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Company 2 with a X% raise _____

Company 4 with a Y% raise _____

Company 7 with a Z% raise _____

Remain at current firm at current pay _____

Worker-Provided Firms

Suppose you planned to move to a new company in the next {one/three/six} months. Would you consider applying to any of these? Please select all that apply.

- ☐ Company 1
- ☐ Company 2
- ☐ Company 3
- ☐ Company 4
- ☐ Company 5
- ☐ Company 6
- ☐ Company 7
- ☐ I would not apply to any of these

“Suppose you planned to move to a new company in the next {one/three/six} months. What are companies that you would consider applying to?

Please list three companies that you would consider applying to and that hire employees in positions like yours (e.g. “Placeholder Inc”). These can be companies without current job vacancies.”

[Fill in Company 1]

[Fill in Company 2]

[Fill in Company 3]

☐ I do not want to answer this question

Worker-Provided Firms

- ▶ Worker-provided firms span a broad range of the German labor market
 - ▶ Total employment in Germany > 6.2 million.
 - ▶ Received more than 190 million page views on employer rating platform Kununu.
- ▶ Nearly 3000 distinct firms—of varying sizes—are mentioned 
 - ▶ Ex: local municipal utility, small manufacturer of agricultural equipment, glass producer

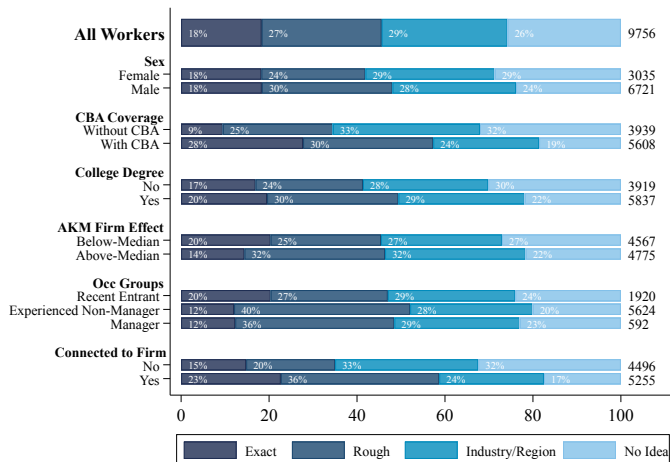
▶ Descriptive Statistics

▶ Comparison with Researcher-Provided

What Do Workers Know About Pay?

1. Many Workers Know Wages at the Time of Application

At the time that you applied, did you know what salary you would earn?



2. Workers Perceive There to Be A Firm Component to Pay

What do you think your gross annual salary would be if you worked at these companies in a position similar to your current one?

	Fraction Identical	Std. Deviation	Max/Min	N
	(1)	(2)	(3)	(4)
<u>A. Researcher-Provided Firms</u>				
Initial Survey	0.26	5673	1.18	3715
Follow-Up Survey	0.30	5294	1.15	3163
<u>B. Worker-Provided Firms</u>				
All Workers	0.25	5863	1.19	4433
All in Same State	0.22	4869	1.18	509
All in Same District	0.26	4701	1.19	173
All in Same Municipality	0.22	5084	1.21	159

Note: We asked 50% of workers to provide expected wages at specific researcher-provided firms. Column 1 reports the share of respondents who indicated the exact same expected pay across all three firms. Columns 2 and 3 report averages across respondents.

2. Workers Perceive There to Be A Firm Component to Pay

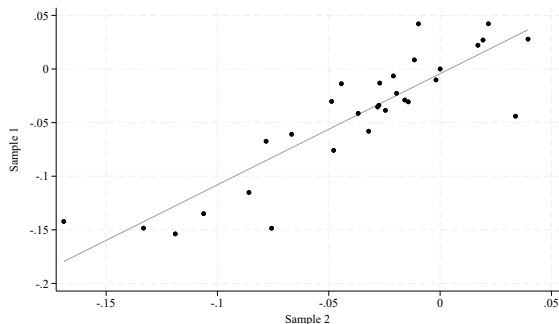
What do you think your gross annual salary would be if you worked at these companies in a position similar to your current one?

$$\log \tilde{w}_{ij} = \alpha_i + \psi_j + \epsilon_{it}. \quad (1)$$

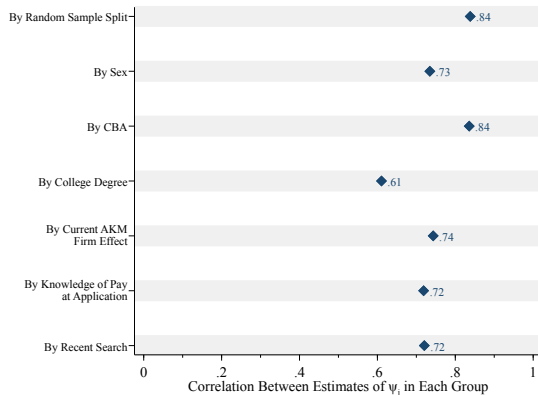
- ▶ Summarize workers' expectations at (randomly-chosen) researcher provided firms ▶ Balance
- ▶ Expected wage premia ψ_j
 - ▶ Identified using within-worker variation in expected salaries
 - ▶ Relative to a “base firm” whose premium is normalized to 0
- ▶ Worker fixed effects α_i
 - ▶ Portion of a worker's salary that they expect to be firm-invariant (across the provided firms)

2. Workers Perceive There to Be A Firm Component to Pay

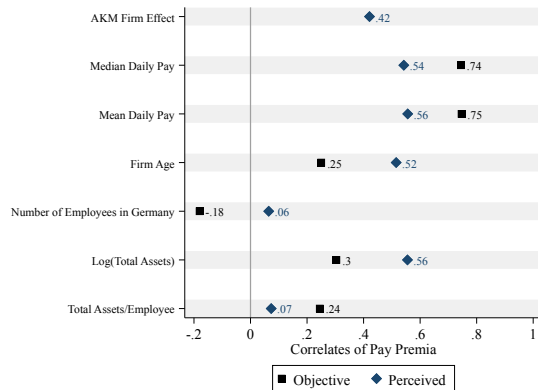
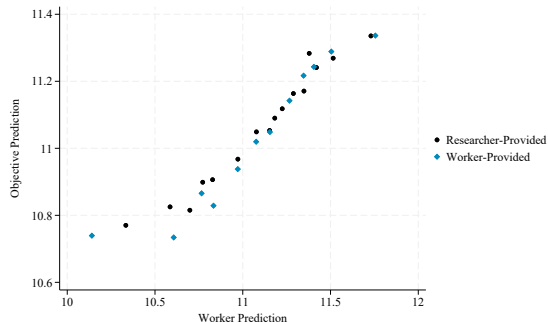
Split-Sample Estimates



Between-Group Correlation in ψ_j



3. Predictions are Correlated with Administrative Data Predictions



Pay Expectations and Consideration

Linking Search to Pay: Within-Worker, Within-Firm Design

$$\text{Consider}_{ij} = \beta \log \tilde{w}_{ij} + X_{ij} + \gamma_j + \lambda_i + \gamma_{t(i)} + \epsilon_{ij}$$

	All Workers			Risk Tolerance		Would be Reluctant to Apply if P(Success) Were Low	
				Low	High	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Own Pay Expectation	0.341*** (0.050)	0.313*** (0.050)	0.309*** (0.050)	0.256*** (0.056)	0.441*** (0.103)	0.297*** (0.084)	0.240*** (0.087)
Distance Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Same-Sector Control	No	No	Yes	Yes	Yes	Yes	Yes
Observations	21272	21272	21272	14967	6305	5392	6507
Number of Workers	6440	6440	6440	4519	1921	1476	1781
Test of equality (p-value)	---	---	---	.116		.64	

Several Designs

▶ Design 1: Link Consideration of Researcher-Provided Firms to:



1. Pay premia workers associate with that firm
2. Observed pay premia
3. Pay policy (Log of mean pay)

Several Designs

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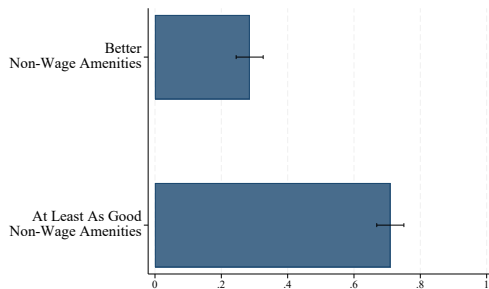
1. Pay premia workers associate with that firm
2. Observed pay premia
3. Pay policy (Log of mean pay)

► Design 2: Link Workers' Free-Text Responses to Pay Premia



Do Workers Believe $\text{cov}(\psi_j, a_j) < 0$?

Compared to a firm that pays 10% above-market wages, one that pays 30% above-market has:



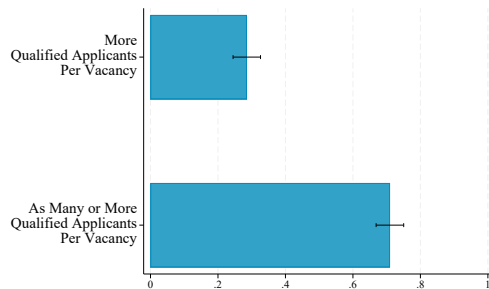
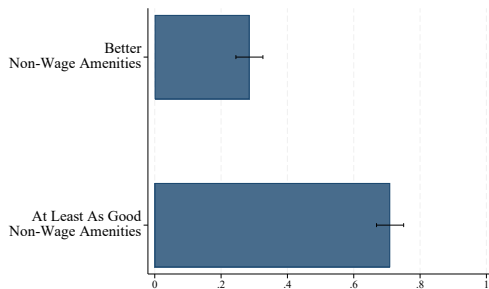
► Question

► Heterogeneity in Beliefs

► Heterogeneity in Beliefs (Figure)

Do Workers Believe in Queuing?

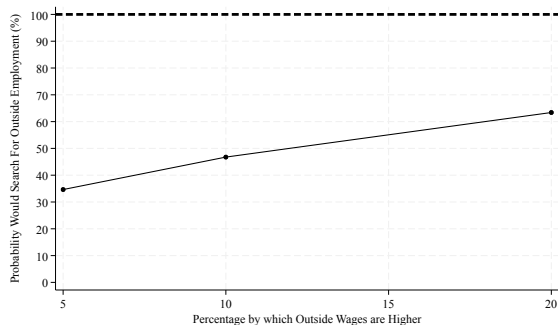
Compared to a firm that pays 10% above-market wages, one that pays 30% above-market has:



Information and Mobility

Not All Workers Will Start Searching

Imagine you were to discover that other companies in your area pay {X%} more than your current employer. How likely is it that you would start applying for jobs at other companies?



Implied Switching Costs

$$u_{ij} = \beta \log w_{ij} + c \cdot 1 \{j(i) \neq j\} + \epsilon_{ij} \quad (2)$$

Main Estimates

	Move to a Researcher- Provided Firm		Move to a Worker-Provided Firm		
	Baseline	Distance Controls	Baseline	Same Commute	Same Growth
	(1)	(2)	(3)	(4)	(5)
Log Raise	6.172*** (0.492)	6.251*** (0.495)	8.112*** (0.824)	12.323*** (1.283)	10.207*** (1.207)
Incumbent	1.132*** (0.074)	0.703*** (0.129)	0.596*** (0.106)	0.771*** (0.134)	0.774*** (0.135)
Observations	29961	29961	17539	8821	8782
Number of Workers	7735	7735	4796	2400	2385
Implied Switching Cost	0.183*** (0.009)	0.112*** (0.019)	0.074*** (0.008)	0.063*** (0.007)	0.076*** (0.008)

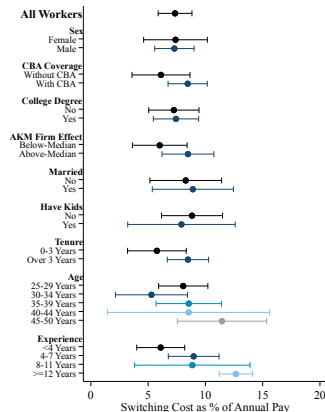
► What Could Information Do?

► Alternative Specifications

Implied Switching Costs

$$u_{ij} = \beta \log w_{ij} + c \cdot 1 \{j(i) \neq j\} + \epsilon_{ij} \quad (3)$$

	Move to a Researcher-Provided Firm		Move to a Worker-Provided Firm		
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Firm-Specific Valuations of Amenities

$$u_{ij} = \beta \log w_{ij} + a_j + (\phi) \cdot 1 \{j(i) \neq j\} + \mu \log d_{ij} + \epsilon_{ij}$$

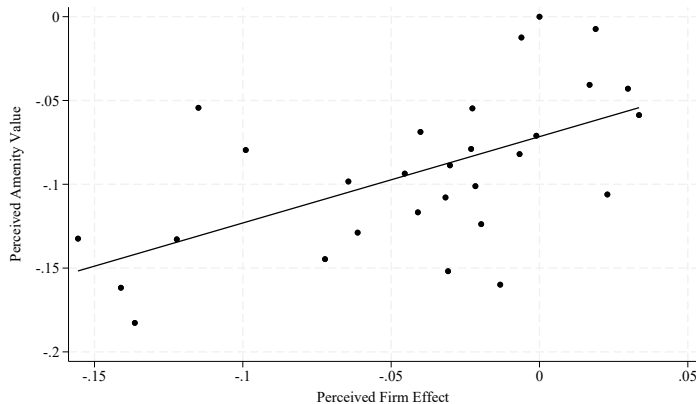
Use workers' preferences over provided firms with randomized raises to identify β and a_j

1. Workers believe firms vary in ex ante rents ($a_j \neq 0$)
2. Workers who would consider applying to the firm have different (and higher) valuations
3. Insiders value firms more highly than outsiders, including those who would consider applying to the firm

Firm-Specific Valuations of Amenities

	Outside Firms Only		All Firms		Consider or Incumbent Only
	(1)	(2)	(3)	(4)	(5)
Log Raise (β)	9.723***				
	(3.131)				
Observations	4217				
Number of Workers (Clusters)	1177				
<u>Test: Ex Ante Firm Effects are Zero</u>					
p-value		<.001			
Chi-Squared Statistic		207.258			
Degrees of Freedom		29			
<u>Test: Ex Ante Effects For Those Who Would and Would Not Apply Are Equal</u>					
p-value					
Chi-Squared Statistic					
Degrees of Freedom					
<u>Test: Ex Post Effects = Ex Ante Effects</u>					
p-value					
Chi-Squared Statistic					
Degrees of Freedom					
<u>Test: Ex Post Effects = Ex Ante Effects For Those Who Would Apply</u>					
p-value					
Chi-Squared Statistic					
Degrees of Freedom					

Firm-Specific Valuations of Amenities



Note: Accounting for the reliability in our estimates, the slope is 0.585 with a standard error of 0.147.

Firm-Specific Valuations of Amenities

	Outside Firms Only		All Firms		Consider or Incumbent Only
	(1)	(2)	(3)	(4)	(5)
Log Raise (β)	9.723***	15.561***			
	(3.131)	(3.235)			
Observations	4217	4217			
Number of Workers (Clusters)	1177	1177			
<u>Test: Ex Ante Firm Effects are Zero</u>					
p-value	<.001	<.001			
Chi-Squared Statistic	207.258	187.988			
Degrees of Freedom	29	29			
<u>Test: Ex Ante Effects For Those Who Would and Would Not Apply Are Equal</u>					
p-value		<.001			
Chi-Squared Statistic		164.417			
Degrees of Freedom		30			
<u>Test: Ex Post Effects = Ex Ante Effects</u>					
p-value					
Chi-Squared Statistic					
Degrees of Freedom					
<u>Test: Ex Post Effects = Ex Ante Effects For Those Who Would Apply</u>					
p-value					
Chi-Squared Statistic					
Degrees of Freedom					

Firm-Specific Valuations of Amenities

	Outside Firms Only		All Firms		Consider or Incumbent Only
	(1)	(2)	(3)	(4)	(5)
Log Raise (β)	9.723*** (3.131)	15.561*** (3.235)	7.008*** (2.461)	9.594*** (2.247)	16.260*** (4.157)
Observations	4217	4217	5671	5671	3001
Number of Workers (Clusters)	1177	1177	1200	1200	1192
<u>Test: Ex Ante Firm Effects are Zero</u>					
p-value	<.001	<.001	<.001	<.001	<.001
Chi-Squared Statistic	207.258	187.988	188.388	131.007	70.891
Degrees of Freedom	29	29	29	29	29
<u>Test: Ex Ante Effects For Those Who Would and Would Not Apply Are Equal</u>					
p-value		<.001		<.001	
Chi-Squared Statistic		164.417		209.519	
Degrees of Freedom		30		30	
<u>Test: Ex Post Effects = Ex Ante Effects</u>					
p-value			<.001	<.001	
Chi-Squared Statistic			7510.727	5633.224	
Degrees of Freedom			17	17	
<u>Test: Ex Post Effects = Ex Ante Effects For Those Who Would Apply</u>					
p-value				<.001	<.001
Chi-Squared Statistic				5592.241	2631.378
Degrees of Freedom				14	12

Conclusions

- ▶ Workers believe firms vary in pay and non-wage values
 - ▶ About half say they had firm-specific pay when they joined their firm
 - ▶ Expectations are correlated with objective values
 - ▶ Amenity valuations are as dispersed as perceived wage premia
- ▶ Workers direct their search on the basis of pay (and amenities)
 - ▶ Firm insiders value amenities more than outsiders
- ▶ Switching costs are large (~7-18% of annual pay)
 - ▶ Information alone is unlikely to have a substantial impact on search or mobility
 - ▶ Importance of other institutions (e.g. unions) for raising wages at the bottom
- ▶ Firms vary in ex post rents

Appendix

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**Deutsche Bahn AG**

als Arbeitgeber

✔ auf kununu aktiv

🇩🇪 [All und weitere](#)**3,8** ★★★★★☆**kununu Score**

13.153 Bewertungen

72%**Weiterempfehlung**

Letzte 2 Jahre

[Übersicht](#)[Bewertungen \(13.153\)](#)[Gehälter \(9.876\)](#)[Jobs \(579\)](#)[Firmenkultur \(3.281\)](#)[Fragen](#)[FOLGEN](#)[ARBEITGEBER BEWERTEN](#)

Mitarbeiter:innen-Zufriedenheit

▼ Karriere & Gehalt	3,3 ★★☆☆☆
▼ Unternehmenskultur	3,6 ★★☆☆☆
▼ Arbeitsumgebung	3,7 ★★☆☆☆
▼ Vielfalt	4,0 ★★★★★

Seit 2007 haben 13.153 Mitarbeiter und Bewerber diesen Arbeitgeber mit durchschnittlich 3,8 Punkten bewertet. Dieser Wert ist höher als der Durchschnitt der Branche [Transport/Verkehr/Logistik](#) (3,3 Punkte).

[Alle 13.153 Bewertungen entdecken](#)

Kenne deinen Wert und verhandle richtig.

9876 Deutsche Bahn Mitarbeiter haben auf kununu bereits für Gehaltstransparenz gesorgt. Finde heraus, wie viel du bei Deutsche Bahn verdienen kannst.

Sortierung: **Anzahl Gehaltsangaben** ▾**Lokomotivführer:in**Ø **40.800 €** brutto/Jahr
704 Gehaltsangaben**Projektmanager:in**Ø **64.600 €** brutto/Jahr
700 Gehaltsangaben**Fahrdienstleiter:in**Ø **40.900 €** brutto/Jahr
474 Gehaltsangaben**Gehaltszufriedenheit****56%****sind mit ihren Gehältern
zufrieden** (basierend auf 8.575
Bewertungen)**Gehalt & Sozialleistungen** **3,5** ★★★★★

Basierend auf 8.575 Bewertungen

Gehaltsinformationen

Aus dem Gehalt wird allzu gern ein Geheimnis gemacht. Gemeinsam mit dir und Millionen anderen kununu Usern können wir für notwendige Gehaltstransparenz sorgen. So findest du den Arbeitgeber, der wirklich zu dir und deinen Vorstellungen passt.

Impact of Randomized Incentives [▶ Back](#)

Initial Survey

	Endorsement	Gift Card		Reminder
	Letter	Level	Binary	
	(1)	(2)	(3)	(4)
Treatment	0.000 (0.002)	-0.000 (0.000)	-0.002 (0.002)	0.040*** (0.001)
Observations	109995	109995	109995	99698

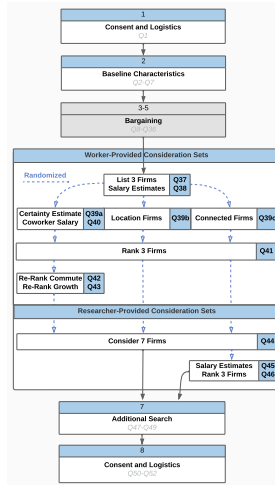
Follow-Up Survey

	Provided an E-Mail			
	No	Yes		
	(1)	(2)	(3)	(4)
Reminder Letter	0.079*** (0.019)			
Initial Letter		0.230*** (0.015)		0.232*** (0.015)
Reminder E-mail			0.070*** (0.016)	0.077*** (0.016)
Constant	0.360*** (0.017)	0.270*** (0.013)	0.393*** (0.014)	0.210*** (0.018)
Observations	3405	5011	5011	5011

Comparison of Respondents and Non-Respondents [▶ Back](#)

	Invited Mean	Linkage Consent			Panel and Linkage		Responded to Follow-Up		
		Mean	Difference Rel. Invited		Mean	Difference Rel. Linked	Mean	Difference Rel. Invited	
	(1)	(2)	(3)		(4)	(5)	(6)	(7)	
<u>Demographics</u>									
Female	0.30 (0.46)	0.32 (0.46)	0.02 (0.00)	***	0.32 (0.47)	0.01 (0.01)	0.31 (0.46)	-0.01 (0.01)	
Age	33.63 (6.59)	33.33 (6.23)	-0.32 (0.06)	***	33.33 (6.14)	-0.02 (0.17)	33.41 (6.16)	0.14 (0.14)	
German Citizen	0.81 (0.39)	0.92 (0.27)	0.12 (0.00)	***	0.92 (0.26)	0.03 (0.01)	0.94 (0.24)	0.02 (0.01)	***
College Education	0.39 (0.49)	0.59 (0.49)	0.22 (0.01)	***	0.60 (0.49)	0.07 (0.01)	0.65 (0.48)	0.07 (0.01)	***
Apprenticeship	0.45 (0.50)	0.33 (0.47)	-0.12 (0.00)		0.32 (0.47)	-0.05 (0.01)	0.29 (0.45)	-0.06 (0.01)	
Daily Earnings	146.03 (60.77)	169.79 (56.71)	25.69 (0.59)	***	170.92 (56.67)	6.61 (1.50)	175.02 (55.27)	7.27 (1.24)	***
Establishments	42705	3556			2983		1457		
Observations	134995	10134			8416		3664		

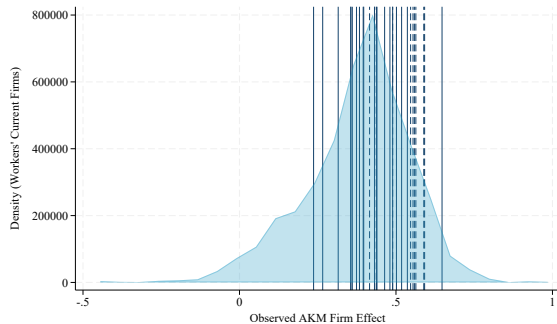
Initial Survey Flow

[▶ Back](#)

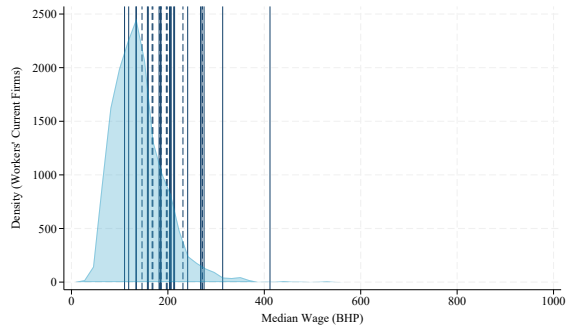
Researcher-Provided Firms vs. Workers' Current Firms

[▶ Back](#)

AKM Firm Effect



Median Pay



Knowledge of Wages at the Time of Application: Question [▶ Back](#)

At the time that you applied, did you know what salary you would earn?

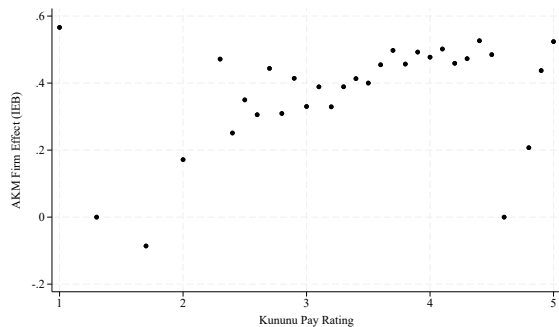
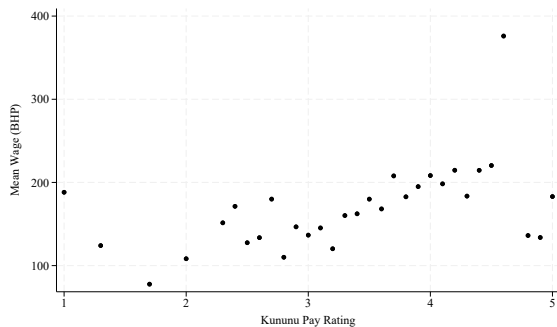
- ▶ I had no or very little idea
- ▶ I only had a rough idea what is paid in my region or sector
- ▶ I had at least a rough idea what this company pays for the position
- ▶ I knew exactly what this company pays for the position

Firm Fixed Effects are Non-Zero Across Specifications [▶ Back](#)

	Sector Fixed Effects		Sector and State Fixed Effects		Sector-State Fixed Effects	
	(1)	(2)	(3)	(4)	(5)	(6)
A. Researcher-Provided Firms						
Adjusted R-Squared	0.858	0.866	0.862	0.866	---	---
Observations	19734	19734	19734	19734	---	---
<u>All Firm Effects are Zero</u>						
Parameters Tested		24		17	---	---
F-Statistic		7		5	---	---
p-value		<.01		<.01	---	---
B. Worker-Provided Firms						
Adjusted R-Squared	0.888	0.907	0.868	0.885	0.871	0.881
Observations	10733	10733	8527	8527	8138	8138
<u>All Firm Effects are Zero</u>						
Parameters Tested		537		386		336
F-Statistic		33792480		6807817		56700000000
p-value		<.01		<.01		<.01

Note: In each column, we regress individuals' log expected earnings on the fixed effects indicated in the headers. The regressions in the even columns add firm fixed effects. We control for worker fixed effects, cluster standard errors at the individual level, and use sampling weights. The F-statistic presented below the observation count comes from testing whether all of the firm dummies are zero.

Comparison of Kununu with Administrative Data [▶ Back](#)



Correlates of Perceived AKM Effects [▶ Back](#)

	Researcher-Provided Firms		Worker-Provided Firms	
	Observed Ψ	Estimated Ψ	Observed Ψ	Estimated Ψ
	(1)	(2)	(3)	(4)
<u>Firm Pay</u>				
Log(Mean Daily Pay) (BHP)	0.57	0.61	0.50	0.27
Log(Median Daily Pay) (BHP)	0.56	0.60	0.48	0.26
AKM FE	1.00	0.37	1.00	0.15
<u>Firm Characteristics</u>				
Age	0.31	0.27	0.01	0.01
Log(Overall Employees)	-0.13	0.42	0.53	0.08
Log(Employees in Germany)	0.03	0.31	-0.06	0.09
Log(Total Assets)	0.30	0.53	0.18	0.21
Log(Fixed Assets)	0.30	0.64	0.13	0.21
<u>Employer Ratings (Kununu)</u>				
Number of reviews	-0.06	-0.21	0.05	0.08
Pct. that would recommend	0.09	0.42	0.14	0.11
Salary rating	0.36	0.56	0.28	0.18
Top salary rating	0.09	0.42	0.14	0.11
	30		715	

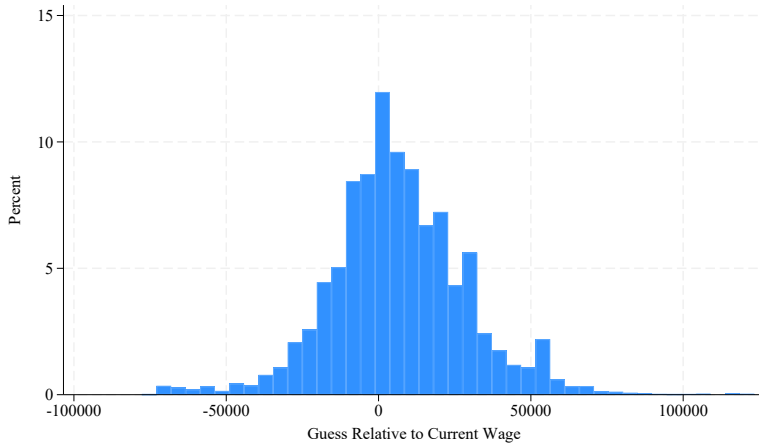
Comparison with AKM Estimates [▶ Back](#)

	Worker Expectations					Objective Predictions
	All Workers	Informed at Application		Recent Search Activity		
		Yes	No	Yes	No	
(1)	(2)	(3)	(4)	(5)	(6)	
<u>Number of Parameters</u>						
Person Effects	5305	2662	2643	3971	1334	5285
Firm Effects	30	30	30	30	30	29
<u>Summary of Parameter Estimates</u>						
Std. Dev. Person Effects	0.365	0.344	0.386	0.358	0.386	0.516
Std. Dev. Firm Effects	0.051	0.049	0.055	0.051	0.053	0.091
RMSE	0.105	0.102	0.108	0.105	0.106	0.336
<u>Addendum</u>						
Std. Dev. Log(Salary)	0.378	0.357	0.401	0.371	0.401	0.328
Variance Log(Salary)	0.143	0.127	0.160	0.138	0.160	0.108
Observations	19431	9739	9692	14580	4851	18754

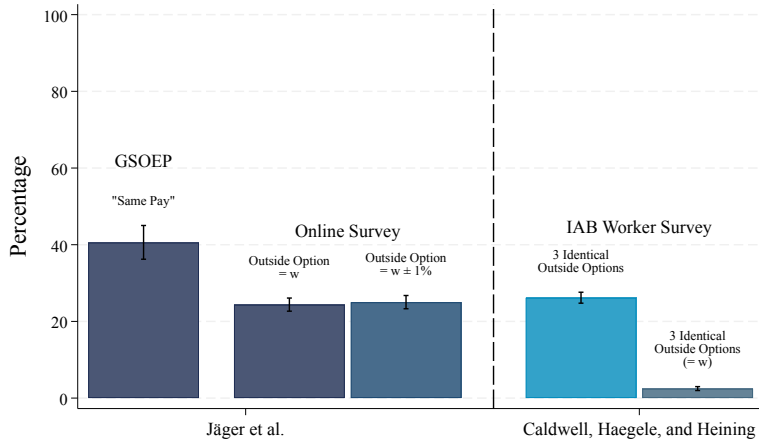
Agreement in Estimates Across Demographic Groups [▶ Back](#)

	Baseline Model		Rank-Ordered Logit	
	Correlation	Test of Equality	Correlation	Test of Equality
		(p-value)		(p-value)
	(1)	(2)	(3)	(4)
Split-Sample	0.88	0.57	0.94	0.65
Sex	0.75	0.02	0.87	0.81
CBA	0.84	0.19	0.91	0.84
College Education	0.65	0.12	0.90	0.52
Current Firm AKM Effect (Split at Median)	0.76	0.01	0.90	0.33
Searched in Past 6 Mo.	0.73	0.02	0.86	0.73
Knew Wages at Application	0.73	0.20	0.92	0.56
Easy to Get a Better Job	0.82	0.38	0.93	0.93
Tenure (Split at 2 Years)	0.86	0.47	0.94	0.45

Anchoring of Workers' Beliefs

[▶ Back](#)[▶ Comparison](#)

Comparison of Anchoring with Jaeger et al. (2025)

[▶ Back](#)[▶ Distribution](#)

Randomization of Researcher-Provided Firms was Successful [▶ Back](#)

	Initial Survey			Follow-Up		
	Firm Group	Firm	Firm Quality	Firm Group	Firm	Firm Quality
	(1)	(2)	(3)	(4)	(5)	(6)
<u>Demographics</u>						
Female	0.375	0.576	0.570	0.094	0.064	0.570
Age	0.094	0.332	0.817	0.886	0.783	0.817
German Citizen	0.334	0.374	0.195	0.591	0.697	0.195
<u>Education</u>						
College	0.526	0.643	0.947	0.005	0.014	0.947
Apprenticeship	0.496	0.714	0.607	0.069	0.146	0.607
<u>Employment and Earnings</u>						
Daily Earnings	0.227	0.189	0.552	0.941	0.971	0.552
Earnings are Censored	0.391	0.764	0.893	0.682	0.822	0.893
Weekly Hours (Survey)	0.085	0.128	0.106	0.451	0.639	0.106
Covered by a CBA (Survey)	0.351	0.785	0.736	0.882	0.965	0.736
<u>Sector</u>						
Manufacturing	0.998	0.999	0.941	0.481	0.704	0.941
Retail	0.628	0.945	0.297	0.813	0.818	0.297
Professional	0.730	0.980	0.785	0.399	0.360	0.785

Note: We perform separate regressions of each covariate (indicated in the row) on the characteristics indicated in the column. Each entry provides the p-value from an F test that all of the included regressor(s) (other than the constant) are equal to zero. P-values are calculated using standard errors clustered at the worker level.

Randomization of Raises was Successful [▶ Back](#)

	Initial Survey	Follow-Up
	(1)	(1)
Number of Employees	0.39	0.31
<u>Sector</u>		
Manufacturing	0.64	0.64
Retail	0.44	0.09
Professional Services	0.71	0.15
Information Services	0.33	0.92
Transportation	0.21	0.34
Finance	0.74	0.27
<u>Other Firm Characteristics</u>		
HQ in Eastern Germany	0.78	0.59
Year of Incorporation	0.79	0.20
<u>Financial Characteristics</u>		
Total Assets per Employee	0.94	0.36
Fixed Assets per Employee	0.85	0.30
<u>Employer listed as</u>		
Largest employer	0.23	0.07
Most popular employer	0.57	0.90
Important Brand	0.40	0.04

Note: This table assesses the randomization of firms to pay offers. We perform separate regressions of each covariate (indicated in the row) on the randomly assigned pay offer. We control for the position of the firm (i.e., whether listed first, second, or third) and cluster standard errors at the worker level. Column 1 provides the p-value from a test that the coefficient on the (randomly assigned) pay offer is zero.

Number of Distinct Firms Mentioned by Workers [▶ Back](#)

	Firms by Number of Mentions		
		Linked to	Linked to
	All	IEB	Orbis
	(1)	(2)	(3)
1 Time	1979	1302	1926
2-9 Times	719	500	645
10-49 Times	111	89	106
50-99 Times	13	11	13
100-249 Times	12	9	11
250+ Times	8	6	7
Total	2842	1917	2708

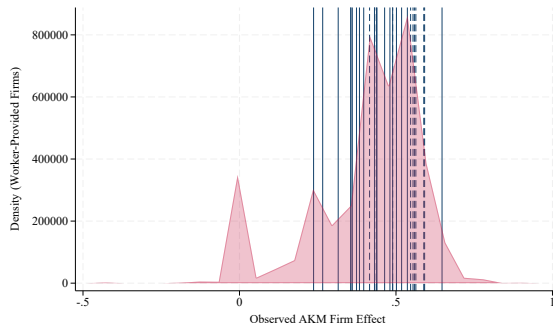
Observed Characteristics of Worker-Provided Firms [▶ Back](#)

	Mean	Std. Dev	N
	(1)	(2)	(3)
<u>Number of Employees</u>			
1-10	0.04	(0.19)	479
11-50	0.07	(0.25)	479
51-200	0.10	(0.29)	479
201-1000	0.22	(0.41)	479
1001-10000	0.34	(0.47)	479
10001+	0.24	(0.43)	479
<u>Sector</u>			
Manufacturing	0.31	(0.46)	565
Retail	0.12	(0.32)	565
Professional Services	0.13	(0.33)	565
Information Services	0.07	(0.26)	565
Finance	0.07	(0.25)	565

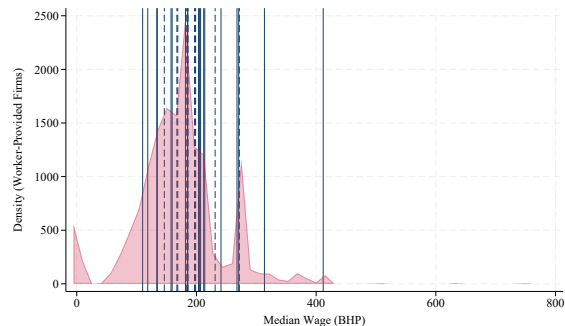
	Mean	Std. Dev	N
	(1)	(2)	(3)
<u>Other Firm Characteristics</u>			
HQ in Eastern Germany	0.07	(0.25)	565
Year of Incorporation	2008	(835)	476
<u>Employer listed as</u>			
Largest employer	0.07	(0.25)	565
Most popular employer	0.07	(0.25)	565
Important brand	0.06	(0.24)	565
<u>Employer ratings</u>			
# page views (in K)	382	(602)	497
# reviews	598	(1090)	497
Top salary rating	0.18	(0.39)	565

Researcher-Provided Firms vs. Worker-Provided Firms [▶ Back](#)

AKM Firm Effect



Median Pay



Characteristics of Researcher-Provided Firms [▶ Back](#)

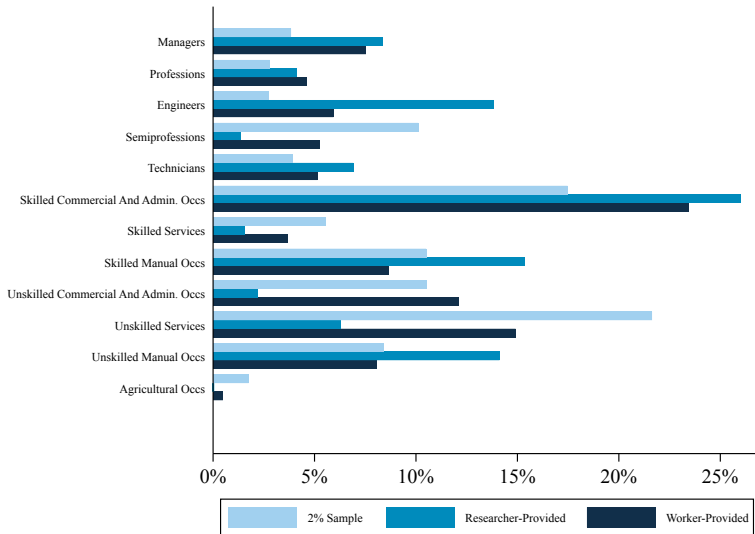
	Mean	Std. Dev	N
	(1)	(2)	(3)
<u>Number of Employees</u>			
1-10	0.00	(0.00)	30
11-50	0.03	(0.18)	30
51-200	0.00	(0.00)	30
201-1000	0.00	(0.00)	30
1001-10000	0.03	(0.18)	30
10001+	0.93	(0.25)	30
<u>Sector</u>			
Manufacturing	0.57	(0.50)	30
Retail	0.07	(0.25)	30
Professional Services	0.10	(0.31)	30
Information Services	0.10	(0.31)	30
Finance	0.10	(0.31)	30

	Mean	Std. Dev	N
	(1)	(2)	(3)
<u>Other Firm Characteristics</u>			
HQ in Eastern Germany	0.07	(0.25)	30
Year of Incorporation	1936	(49)	30
<u>Employer listed as</u>			
Largest employer	0.63	(0.49)	30
Most popular employer	0.53	(0.51)	30
Important brand	0.50	(0.51)	30
<u>Employer ratings</u>			
# page views (in K)	1305	(1216)	30
# reviews	2339	(2796)	30
Top salary rating	0.40	(0.50)	30

Researcher-Provided vs Worker-Provided Firms [▶ Back](#)

	Researcher-Provided Firms			Worker-Provided Firms		
	Mean	Std. Dev	N	Mean	Std. Dev	N
	(1)	(2)	(3)	(4)	(5)	(6)
<u>Number of Employees</u>						
1-10	0.00	(0.00)	30	0.04	(0.19)	479
11-50	0.03	(0.18)	30	0.07	(0.25)	479
51-200	0.00	(0.00)	30	0.10	(0.29)	479
201-1000	0.00	(0.00)	30	0.22	(0.41)	479
1001-10000	0.03	(0.18)	30	0.34	(0.47)	479
10001+	0.93	(0.25)	30	0.24	(0.43)	479
<u>Sector</u>						
Manufacturing	0.57	(0.50)	30	0.31	(0.46)	565
Retail	0.07	(0.25)	30	0.12	(0.32)	565
Professional Services	0.10	(0.31)	30	0.13	(0.33)	565
Information Services	0.10	(0.31)	30	0.07	(0.26)	565
Finance	0.10	(0.31)	30	0.07	(0.25)	565
<u>Other Firm Characteristics</u>						
HQ in Eastern Germany	0.07	(0.25)	30	0.07	(0.25)	565
Year of Incorporation	1936	(49)	30	2008	(835)	476
<u>Employer listed as</u>						
Largest employer	0.63	(0.49)	30	0.07	(0.25)	565
Most popular employer	0.53	(0.51)	30	0.07	(0.25)	565
Important brand	0.50	(0.51)	30	0.06	(0.24)	565
<u>Employer ratings</u>						
# page views (in K)	1305	(1216)	30	382	(602)	497
# reviews	2339	(2796)	30	598	(1090)	497
Top salary rating	0.40	(0.50)	30	0.18	(0.39)	565
% recommended	74	(13)	30	70	(17)	486

Occupational Distribution

[▶ Back](#)

Linking Search to Pay: Researcher-Provided Firms [▶ Back](#)

$$\text{Consider}_{ij} = \beta^{RP} \text{Pay}_j + \lambda_i + X_{ij} + \epsilon_{ij}$$

	Stated Consideration			Free-Text Responses		
	(1)	(2)	(3)	(4)	(5)	(6)
Mean of Dependent Variable		0.254			0.017	
A. Perceived Firm Effect (Split-Sample)						
Firm Premium (Split-Sample)	0.892*** (0.110)	0.983*** (0.127)	0.986*** (0.127)	0.097*** (0.011)	0.099*** (0.013)	0.101*** (0.013)
Observations	89742	89742	89742	224388	224388	224388
Number of Workers	9756	9756	9756	9756	9756	9756
B. Observed Firm Effect						
Firm Premium (Observed)	0.174*** (0.033)	0.165*** (0.033)	0.173*** (0.035)	0.014*** (0.005)	0.014*** (0.005)	0.014*** (0.005)
Observations	89258	89258	89258	214632	214632	214632
Number of Workers	9756	9756	9756	9756	9756	9756
C. Observed Log(Mean Daily Pay)						
Firm Mean Daily Pay	0.086*** (0.011)	0.093*** (0.011)	0.093*** (0.011)	0.008*** (0.002)	0.008*** (0.002)	0.008*** (0.002)
Observations	89258	89258	89258	214632	214632	214632
Number of Workers	9756	9756	9756	9756	9756	9756
Firm Characteristics						
	Size	Size, Brand Recognition	Size, Brand Recognition, CBA	Size	Size, Brand Recognition	Size, Brand Recognition, CBA
Fixed Effects	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector

Note: The outcome variable in Columns 1-3 is an indicator for whether the worker checked the box indicating they would consider applying to the firm if they wanted to switch firms. Regressions use sampling weights. Standard errors are clustered at the worker level.

Linking Search to Pay: Worker-Provided Firms [▶ Back](#)

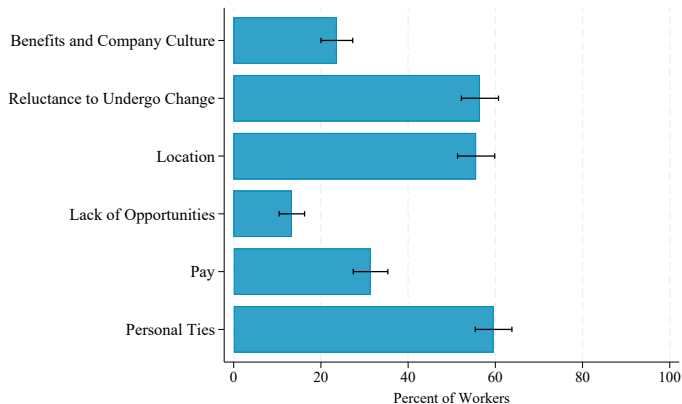
$$\text{List Firm}_{ij} = \beta^{WP} \text{Pay}_j + \lambda_i + X_{ij} + \epsilon_{ij}$$

	Stated Consideration			Free-Text Responses		
	(1)	(2)	(3)	(4)	(5)	(6)
Mean of Dependent Variable		0.254			0.017	
A. Perceived Firm Effect (Split-Sample)						
Firm Premium (Split-Sample)	0.892*** (0.110)	0.983*** (0.127)	0.986*** (0.127)	0.097*** (0.011)	0.099*** (0.013)	0.101*** (0.013)
Observations	89742	89742	89742	224388	224388	224388
Number of Workers	9756	9756	9756	9756	9756	9756
B. Observed Firm Effect						
Firm Premium (Observed)	0.174*** (0.033)	0.165*** (0.033)	0.173*** (0.035)	0.014*** (0.005)	0.014*** (0.005)	0.014*** (0.005)
Observations	89258	89258	89258	214632	214632	214632
Number of Workers	9756	9756	9756	9756	9756	9756
C. Observed Log(Mean Daily Pay)						
Firm Mean Daily Pay	0.086*** (0.011)	0.093*** (0.011)	0.093*** (0.011)	0.008*** (0.002)	0.008*** (0.002)	0.008*** (0.002)
Observations	89258	89258	89258	214632	214632	214632
Number of Workers	9756	9756	9756	9756	9756	9756
Firm Characteristics						
	Size	Size, Brand Recognition	Size, Brand Recognition, CBA	Size	Size, Brand Recognition	Size, Brand Recognition, CBA
Fixed Effects	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector

Note: The outcome variable in Columns 4-6 is an indicator for whether the worker listed each of the firms in the researcher-provided not random order.

Why Don't Workers Want to Move? [▶ Back](#)

We asked workers to select the two main reasons employees are reluctant to switch jobs.

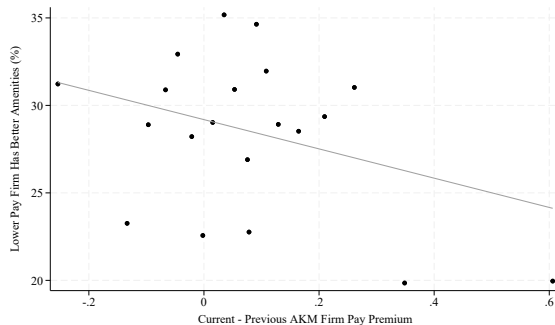
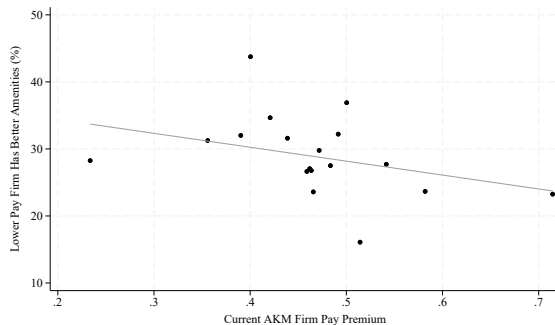


Heterogeneity in the Belief That $\text{cov}(\psi_j, a_j) < 0$

[▶ Back](#)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Firm Pay Premium	-0.385*	-0.388**	-0.418**	-0.515**				
	(0.220)	(0.173)	(0.209)	(0.215)				
Current - Former Firm Pay Premium					-0.153	-0.161*	-0.258**	-0.287***
					(0.101)	(0.097)	(0.120)	(0.111)
Log(Hours)		-0.152	0.093	0.155		-0.258	0.158	0.236
		(0.213)	(0.109)	(0.112)		(0.269)	(0.164)	(0.166)
Log(Wage)		0.015	0.124**	0.167***		0.009	0.074	0.125*
		(0.070)	(0.058)	(0.059)		(0.072)	(0.070)	(0.066)
Female				0.158***				0.150**
				(0.056)				(0.061)
College Degree				-0.083				-0.103
				(0.059)				(0.067)
Experience				0.001				-0.003
				(0.004)				(0.004)
Constant	0.422***	0.904	-0.488	-0.910**	0.264***	1.161	-0.655	-1.165**
	(0.109)	(0.928)	(0.397)	(0.415)	(0.040)	(1.193)	(0.554)	(0.571)
Sector Fixed Effects	---	---	Yes	Yes	---	---	Yes	Yes
Adjusted R-Squared	0.023	0.028	0.283	0.303	0.006	0.023	0.266	0.286
Observations	1642	1642	1642	1642	1384	1384	1384	1384

Heterogeneity in the Belief That $\text{cov}(\psi_j, a_j) < 0$

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Robustness to Additional Specifications [► Back](#)

	Alternative Specifications of Distance			Alternative Samples		Alternative Weighting Schemes	
	Quadratic in Distance	Direct Distance	Closest Establishment	Initial Survey Only	Follow-Up Only	Unweighted	Population Weights
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Own-Pay Expectation	0.312*** (0.050)	0.309*** (0.050)	0.331*** (0.050)	0.427*** (0.122)	0.249 (0.174)	0.225*** (0.026)	0.255*** (0.085)
Observations	21272	21272	21272	15095	5980	21272	21236
Number of Workers (Clusters)	6440	6440	6440	0.396	0.309	0.330	0.335
Mean of Outcome	0.31	0.31	0.31	0.35	0.24	0.33	0.32
Implied Elasticity	1.004*** (0.162)	0.996*** (0.161)	1.067*** (0.161)	1.214*** (0.347)	1.052 (0.736)	0.677*** (0.079)	0.788*** (0.261)

What Workers Believe About High Wage Firms [▶ Back](#)

Suppose you are comparing job opportunities at two different companies: Company 1 pays 10% above the market average and Company 2 pays 30% above the market average.

1. Which company do you think attracts more qualified applicants per opening?
 - ▶ Company 1
 - ▶ Company 2
 - ▶ Both attract the same number of applicants
2. Which company do you think provides better non-wage amenities (e.g., home office, childcare subsidy)?
 - ▶ Company 1
 - ▶ Company 2
 - ▶ Both provide the same non-wage amenities

Reynolds (1951) [▶ Back](#)

*“These results confirm the prevalent impression that workers are poorly informed about job opportunities. Moreover, it is doubtful how far the situation can be altered by collecting and disseminating additional job information. **The basic difficulty is that satisfactorily employed workers are almost entirely uninterested in employment conditions in other companies. This lack of interest is an even more serious obstacle than the difficulty of compiling accurate job information [emphasis added]**”*
–Reynolds (1951)

What Could Information Do? [▶ Back](#)

	All Workers Are Uninformed	Did Not Know Pay at Application	Difficult to Get a Better Job	Provide Uniform Pay
	(1)	(2)	(3)	(4)
Percent Informed	0 ---	49.826*** (0.854)	44.107*** (0.897)	77.288*** (1.131)
A. Gap Between Median and Current Firm				
P(Search)	17.606*** (2.504)	4.867*** (0.761)	4.356*** (0.758)	2.048*** (0.472)
P(Move to a Preferred Firm)	11.837*** (2.904)	2.784*** (0.789)	2.579*** (0.675)	1.486*** (0.453)
P(Move to a Random Firm)	1.127 (1.761)	0.837* (0.475)	0.963** (0.454)	0.741** (0.294)
B. 5% Wage Gain				
P(Search)	9.610*** (1.263)	4.599*** (0.633)	4.996*** (0.748)	2.593*** (0.482)
P(Move) to a Preferred Firm	6.977*** (1.883)	3.845*** (0.942)	4.009*** (0.902)	2.450*** (0.676)
P(Move) to a Random Firm	2.473** (1.041)	0.947* (0.526)	1.156** (0.587)	0.926** (0.373)

Note: In each column of this table, we estimate the impact of information with a different definition of being uninformed in each column. The sample includes workers who work at firms with below-median pay premia. In Panel A, we estimate the impact if we informed workers of the difference between the median worker's firm premium and their own firm's. In Panel B, we perform an analogous analysis if all workers were told they could receive 5% more at outside firms. Coefficients and standard errors are estimated via bootstrapping.

Alternative Specifications for Switching Costs [▶ Back](#)

	Alternative Specifications of Distance				Alternative Samples		Alternative Weighting	
	Log	Quadratic	Direct	Closest	Initial			
	Distance	in Distance	Distance	Establishment	Survey Only	Follow-Up Only	Unweighted	Population Weights
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Researcher-Provided Firms								
Implied Switching Cost	0.112*** (0.019)	0.123*** (0.011)	0.113*** (0.019)	0.167*** (0.011)	0.142*** (0.025)	0.083*** (0.023)	0.127*** (0.007)	0.094*** (0.022)
Observations	29961	29961	29961	29961	16594	13367	29961	29961
Number of Workers	7735	7735	7735	7735	4322	2351	7735	7735
B. Worker-Provided Firms								
Implied Switching Cost	0.079*** (0.010)	0.078*** (0.008)	0.066*** (0.009)	0.072*** (0.008)	---	---	0.099*** (0.005)	0.059*** (0.010)
Observations	15259	15259	17539	17539			15259	15259
Number of Workers	4784	4784	4796	4796			4784	4784